

Jonas Ngnawé

Quebec City, Canada

ngnawejonas@gmail.com • ngnawejonas.github.io • linkedin.com/in/jonasngnawé • Google Scholar

Education

Université Laval, Mila - Quebec AI Institute, Stanford University (VSR) <i>PhD in Computer Science: Adversarial Machine Learning</i>	Quebec City, Canada 2021–2027 (expected)
AIMS African Master's of Machine Intelligence <i>MSc in Machine Learning</i> Funded by Meta and Google; taught by world top researchers .	Kigali, Rwanda 2018–2019
African Institute for Mathematical Sciences (AIMS) <i>MSc in Mathematical Sciences</i>	Limbe, Cameroon 2016–2017
École Polytechnique Yaoundé <i>Engineering Degree (MSc equivalent), Computer Engineering</i> Five-year integrated engineering programme.	Yaoundé, Cameroon 2011–2016
Summer Schools: Machine Learning Summer School (MLSS) , London 2019; Deep Learning & Reinforcement Learning Summer School (DLRL) , Edmonton 2025.	

Publications

Peer-reviewed conference papers

- RobustGenBench: A Benchmark for Robust Generalization to Adversarial and Common Perturbations, with Applications to Vision and Vision-Enabled Large Language Models. *Heuillet, M., Ngnawé, J., Bhagwatkar, R., Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A.* **NeurIPS 2026**, Evaluations and Datasets Track.
- Robust Fine-Tuning from Non-Robust Pretrained Models: Mitigating Suboptimal Transfer with Epsilon-Scheduling. *Ngnawé, J., Heuillet, M., Sahoo, S., Pequignot, Y., Ahmad, O., Durand, A., Précioso, F., & Gagné, C.* **ICLR 2026** (earlier version: NeurIPS 2025 Workshop on Reliable ML). [\[OpenReview\]](#)
- Detecting Brittle Decisions for Free: Leveraging Margin Consistency in Deep Robust Classifiers. *Ngnawé, J., Sahoo, S., Pequignot, Y., Précioso, F., & Gagné, C.* **NeurIPS 2024**. [\[Proceedings\]](#)
- A Layer Selection Approach to Test Time Adaptation. *Sahoo, S., ElAraby, M., Ngnawé, J., Pequignot, Y., Précioso, F., & Gagné, C.* **AAAI 2025**. [\[Proceedings\]](#)

Workshop papers

- A Guide to Robust Generalization: The Impact of Architecture, Pre-training, and Optimization Strategy. *Heuillet, M., Bhagwatkar, R., Ngnawé, J., Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A.* NeurIPS 2025 Workshop on Reliable ML. [\[arXiv\]](#)
- Robustmix: Improving Robustness by Regularizing the Frequency Bias of Deep Networks. *Ngnawé, J., Njifon, M. A., Heek, J., & Dauphin, Y.* NeurIPS 2022 Workshop on Distribution Shifts. [\[OpenReview\]](#)

Under review

- Mind the Spike: Mechanisms and Brittleness of Visual Massive Activations in Large Vision–Language Models. *Ngnawé, J., Pequignot, Y., Sahoo, S., Gagné, C., Précioso, F., & Koyejo, S.* [\[arXiv\]](#)
- Margin-based Progress Measures for Grokking. *Ngnawé, J., Notsawo Jr, P.*
- A Stochastic–Geometric Theory of Scaling Laws in Grokking. *Luo, R., Gagné, C., Ngnawé, J., Ullah, I. and Morrissey, K.* [\[arXiv\]](#)

AI/ML Work Experience

Vector Institute (FastLane MLA Program) <i>Machine Learning Associate</i>	Toronto, Canada May – Aug 2026
○ Optimized LLM inference for AutoInference.ai: vLLM serving config to increase goodput and reducing latency.	

Stanford Artificial Intelligence Lab (SAIL)

Visiting Student Researcher

Palo Alto, USA

Oct 2025 – March 2026

- Safety, robustness and uncertainty estimation in multimodal vision–language models.

Google Brain

Google AI Resident

Accra, Ghana

Oct 2019 – Sep 2021

- Developed robust training methods for deep networks; frequency-bias regularization work published at the NeurIPS 2022 Distribution Shifts workshop.
- Compressed deep networks via quantization and pruning: Mixed-precision quantization ($\{4, 8\}$ -bit quantization, with 0-bit for sparsity to retain high accuracy on vision models).

Earlier Software Engineering Experience

GroupOne Holding Company

Computer Engineer

Limbe, Cameroon

Jan – Aug 2018

Web development and fuel-consumption prediction; industrial maintenance of Honeywell Experion refinery systems. *JavaScript, PHP, MySQL*.

Freelance

Software Developer

Cameroon

2017–2018

Digitized administrative procedures for Cameroonian ministries; built and maintained WooCommerce plugins. *Maarch RM, PHP, Java, MySQL, Git*.

Bolloré Africa Logistics

Software Developer Intern

Douala, Cameroon

Feb – Aug 2016

Automated review of information-system access rights and built anomaly detection for abnormal privileges. *Talend, QlikView, C#, ASP.NET, SQL Server*.

Talks

Host, Kmer AI Talks.

2026: “Robust Fine-Tuning from Non-Robust Pretrained Models” — [Galsen AI \(2026\)](#).

2024–2025: “Detecting Brittle Decisions for Free” — [Kmer AI \(Jun 2024\)](#); [Galsen AI \(Jan 2025\)](#).

2024: Panelist, “AI Regulation & Fairness in the Generative AI Era,” Black in AI Workshop @ NeurIPS.

2022: Keynote, “Robustness in AI,” Deep Learning IndabaX-Cameroon (Dec 2022).

Service

Reviewer: NeurIPS, ICLR, AAAI; IEEE Transactions on Image Processing.

Algorithmic Fairness workshops @ NeurIPS 2021–2024; Montreal AI Symposium (2022, 2024); Black in AI Workshop (2022–2024); Deep Learning Indaba / Pan-African ML Conference (2023).

Organizing: Interpolate Workshop @ NeurIPS 2022 (First Workshop on Interpolation Regularizers and Beyond), organizing committee.

Co-organizer, Machine Learning Projects Competition (MLPC) 2020–2022, Polytechnique Yaoundé, \$2,000 sponsorship.

Co-organizer: two-week introductory ML course (Sep 2019), Polytechnique Yaoundé; first Deep Learning IndabaX Rwanda (Apr 2019), Kigali.

Co-founder, Kmer-AI – association promoting AI and ML in Cameroon (since 2020).

AI/ML Skills

ML Frameworks: PyTorch, TensorFlow, Scikit-Learn

MLOps Tools: Git/GitHub, Weights & Biases, Docker, Slurm, GCP, AWS, vLLM, Alperf